

Holistic Data Preparer

1. Data Analysis

Data Analysis is the process of collecting, cleaning, transforming, and interpreting data to extract useful insights. It helps in understanding patterns, trends, and relationships in data, which supports decision-making. In this project, data analysis is used to understand customer financial behavior and predict loan default risk.

2. Data Acquisition

Data acquisition refers to collecting data from multiple sources. In this project, data is collected from:

- CSV files for main customer data
- JSON files for additional metadata
- SQL databases for repayment history
- APIs for external data

Combining multiple sources improves data richness and model accuracy.

3. Data Cleaning

Data cleaning is the process of handling inconsistencies, errors, and missing values in the dataset. It ensures that the dataset is accurate and reliable. Common steps include:

- Removing duplicates
- Handling missing values
- Correcting data types

Clean data is essential for building effective machine learning models.

4. Missing Value Handling

Missing values occur when data is incomplete. Several techniques are used:

- Mean/Median Imputation: Replace missing numerical values with average or median
- Most Frequent Imputation: Replace missing categorical values with the most common value
- Random Sample Imputation: Replace missing values using random existing values
- Missing Indicator: Create a new column indicating missing data
- KNN Imputer: Uses nearest neighbors to estimate missing values
- MICE (Multiple Imputation): Uses regression-based iterative imputation

- Complete Case Analysis: Removes rows with missing values

Proper handling prevents bias and improves model performance.

5. Outlier Handling

Outliers are extreme values that differ significantly from other observations. They can distort analysis and model performance.

Methods used:

- Z Score Method: Detects outliers based on standard deviation
- IQR Method: Uses quartiles to detect extreme values
- Percentile Method: Caps values at specific percentiles
- Winsorization: Limits extreme values without removing them

Handling outliers ensures stability and accuracy of models.

6. Feature Engineering

Feature engineering is the process of creating new variables from existing data to improve model performance.

Examples in this project:

- Debt-to-Income Ratio
- Average Monthly Transactions
- Spending-to-Income Ratio

These features help the model better understand financial risk.

7. Handling Variable Types

Datasets often contain different types of variables:

- Numerical variables (continuous values)
- Categorical variables (labels or categories)
- Date/time variables

Date variables are transformed into useful components such as year, month, day, and weekday to extract patterns.

8. Encoding Categorical Variables

Machine learning models require numerical input, so categorical data must be converted.

Methods used:

- Ordinal Encoding: Assigns ordered numbers to categories
- Label Encoding: Converts categories into numeric labels
- One-Hot Encoding: Creates binary columns for each category

Encoding ensures compatibility with machine learning algorithms.

9. Numerical Feature Encoding

Numerical features are sometimes transformed into categories:

- Binning: Divides values into groups
- Binarization: Converts values into 0 or 1 based on a threshold
- Quantile Binning: Divides data into equal-sized groups
- K-Means Binning: Uses clustering to create groups

These methods simplify patterns and improve model interpretability.

10. Feature Scaling

Feature scaling ensures that all numerical features are on a similar scale, which improves model performance.

Methods used:

- Standardization (Z-score): Centers data around mean with unit variance
- Normalization: Scales data to unit length
- Min-Max Scaling: Scales data between 0 and 1
- MaxAbs Scaling: Scales using maximum absolute value
- Robust Scaling: Uses median and IQR, resistant to outliers

Scaling is important for algorithms like KNN and gradient-based models.

11. Data Transformation

Transformations are applied to reduce skewness and improve data distribution.

Methods used:

- Log Transformation: Reduces right skewness
- Square Root Transformation: Stabilizes variance
- Reciprocal Transformation: Reduces impact of large values

- Power Transformation (Box-Cox, Yeo-Johnson): Makes data more normal

These improve model learning and accuracy.

12. Column Transformer

Column Transformer allows applying different preprocessing steps to different columns in a dataset. For example:

- Scaling numerical features
- Encoding categorical features

It helps build efficient and organized preprocessing pipelines.

13. Final Dataset Preparation

After all preprocessing steps:

- Missing values are handled
- Outliers are treated
- Features are encoded and scaled
- New features are created

The final dataset is clean, consistent, and ready for machine learning modeling.

Conclusion

This project demonstrates a complete data preprocessing pipeline including data cleaning, transformation, feature engineering, and scaling. Each step improves data quality and prepares it for accurate and reliable machine learning predictions in credit risk analysis.